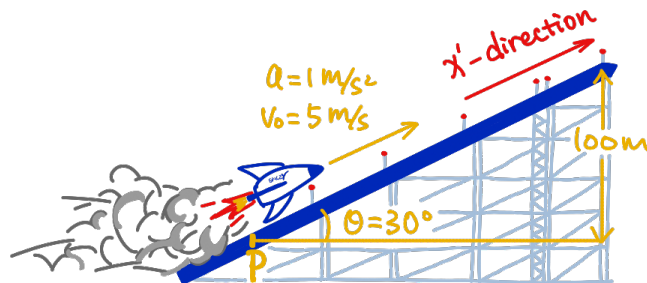


Physics 5A Discussion (Week 2)

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Problem 1: Jinyan is launching a rocket using a ramp at an angle of $\theta = 30^\circ$ with respect to the horizontal. When the rocket reaches point P as shown, it has an instantaneous velocity $v_0 = 5 \text{ m/s}$ in the x' -direction, and its acceleration in the x' -direction is kept constant at 1 m/s^2 . How long does it take for the rocket to reach a vertical displacement of $h = 100 \text{ m}$ (measured with respect to the height of point P)?

This is quite a complicated problem, but we can break it down into a few steps:

Step 1: Looking at the triangle shown, how far does the rocket need to travel in the x' -direction (on the ramp) to reach $h = 100 \text{ m}$?

Step 2: Draw the acceleration vs. time graph, with the acceleration in the x' -direction on the vertical axis, and time on the horizontal axis. (*Hint: As mentioned in the problem, acceleration is kept constant over time.*)

Step 3: From step 2, draw the velocity vs. time graph for the first 20 seconds, and write down the mathematical expression for the instantaneous velocity as a function of time. (*Hint: The change in velocity is the “area” under the acceleration curve.*)

Step 4: From step 3, deduce the shape of curve in the position vs. time graph. Is it a straight line, parabola, or an exponential curve? (*Hint: The slope of your position curve at each time instant should be the value of your velocity curve at the same time instant.*)

Step 5: Again using your velocity vs. time graph, how far has the rocket traveled in the x' -direction in the first 20 seconds? (*Hint: Use the "area" under the velocity curve to find the displacement.*)

Step 6: Recall the distance calculated in step 1, how long does the rocket take to travel such a distance? (*Hint: Try an arbitrary final time t_f and repeat from step 3. You will need the quadratic formula when computing your final answer.*)

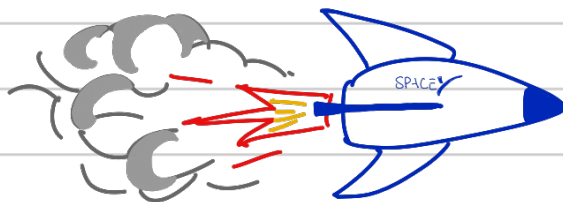
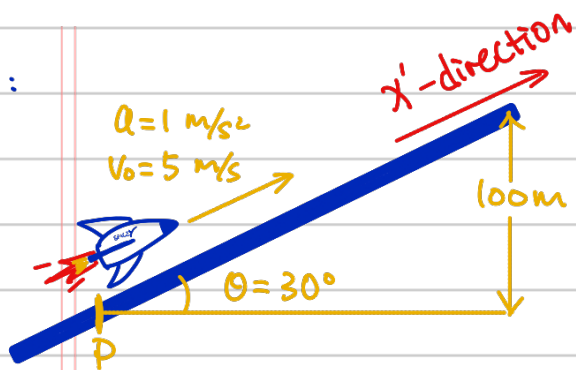
Step 7: Now you have seen that *position curve looks like a parabola if we have a constant acceleration over time*, this result leads to the kinematic equation that you will see very soon during lecture:

$$x = \frac{1}{2}at^2 + v_0t + x_0 ,$$

where x_0 is your initial position at time $t = 0$ second, and x is your final position at a generic time t . Using this formula, find the time t for the rocket to travel from $x_0 = 0$ m to $x =$ (your result in step 1) in the x' -direction. (*Hint: You will need the quadratic formula. The result you get here should agree with the one you found in step 6.*)

Problem 2: For the rocket in problem 1, how far has the rocket traveled in the horizontal direction (x -direction) when it reaches $h = 100$ m?

Setup:



Step 1.

the rocket is travelling on the incline (hypotenuse)
We want it to go up by 100 m (opposite side)
→ trigonometry!!

$$\sin(30^\circ) = \frac{100 \text{ m}}{\Delta x'}$$

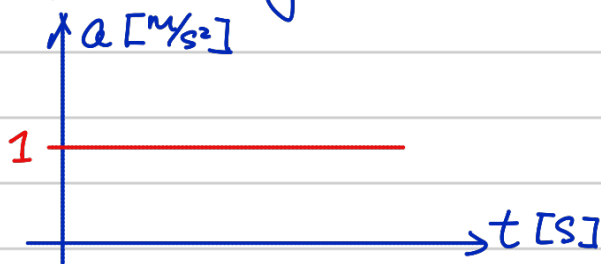
$$\sin(30^\circ) = \frac{1}{2}$$

$\Delta x'$ is displacement on the incline

$$\text{so } \boxed{\Delta x' = 200 \text{ m}}$$

Step 2

acceleration is kept constant $a = 1 \text{ m/s}^2$ over time
→ flat straight line



Step 3

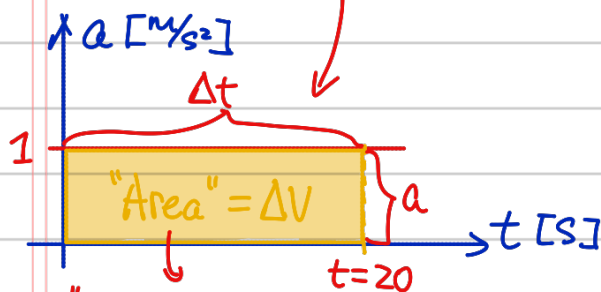
remember $a = \lim_{\Delta t \rightarrow 0} \frac{\Delta V}{\Delta t}$ if a is constant, ignore "lim"
→ $\Delta V = a \cdot \Delta t$ for constant acceleration.

$$\text{Change in velocity } \Delta V = V_{\text{final}} - V_{\text{initial}}$$

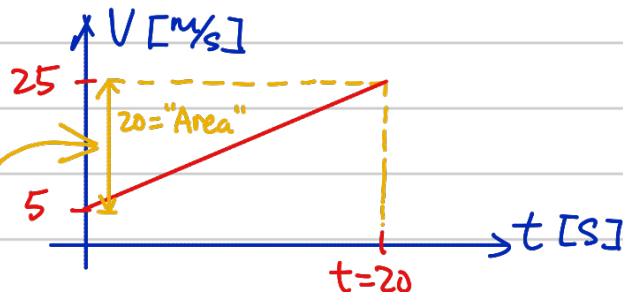
you pick a final time and
this is instantaneous velocity
at that time. In this case
we pick $t_{\text{final}} = 20 \text{ s}$

this is $v_0 = 5 \text{ m/s}$ at $t = 0 \text{ s}$
(given in the problem)

$$\text{so } \Delta V = a \cdot \Delta t = (1 \text{ m/s}) \cdot (20 \text{ s} - 0 \text{ s}) = 20 \text{ m/s}$$

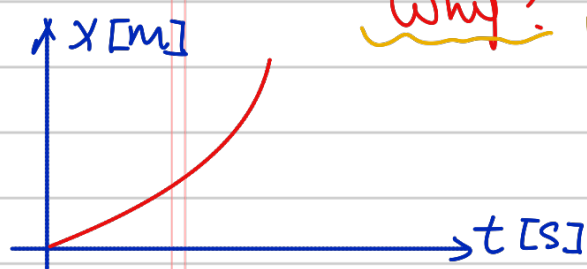


"Area" = 20 m/s



Step 4

As you have seen many times —
when velocity is a tilted straight line, position curve
should be a **parabola**, more specifically, v is growing
in this problem, so the parabola is **concave-up**.
Since v is always larger than 0 in this problem,
we only have the **right-half** of the parabola.



Why?

because rocket is moving faster
and faster to the right
→ x grows faster and faster
over time, never decreases

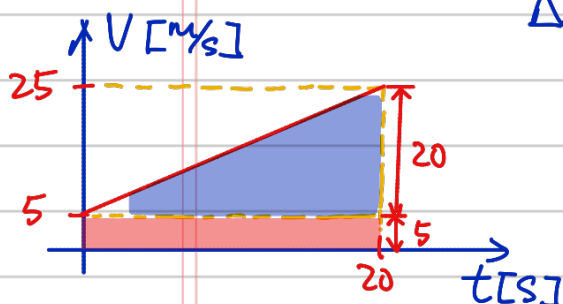
Step 5

We want to find
this number → ?

to do this, recall what we have

$$\Delta x'_{0 \rightarrow 20} = \text{"Area" under } v \text{ from } 0 \text{ s to } 20 \text{ s}$$

displacement from $t=0$ to $t=20$ s
(or change in position)



$$\Delta x'_{0 \rightarrow 20} = \text{"Area" under } v \text{ from } 0 \text{ s to } 20 \text{ s}$$

$$= \text{Area of } \triangle + \text{Area of } \square$$

$$= (20 \text{ s}) \cdot (25 \text{ m/s} - 5 \text{ m/s}) / 2 + (20 \text{ s}) \cdot (5 \text{ m/s})$$

$$= 200 \text{ m} + 100 \text{ m} = \boxed{300 \text{ m}}$$

So the rocket has traveled 300 m in 20 seconds, which is more
than the 200 m that we want for it to go up 100 m vertically.
→ it takes less than 20 s for it to travel 200 m on the ramp

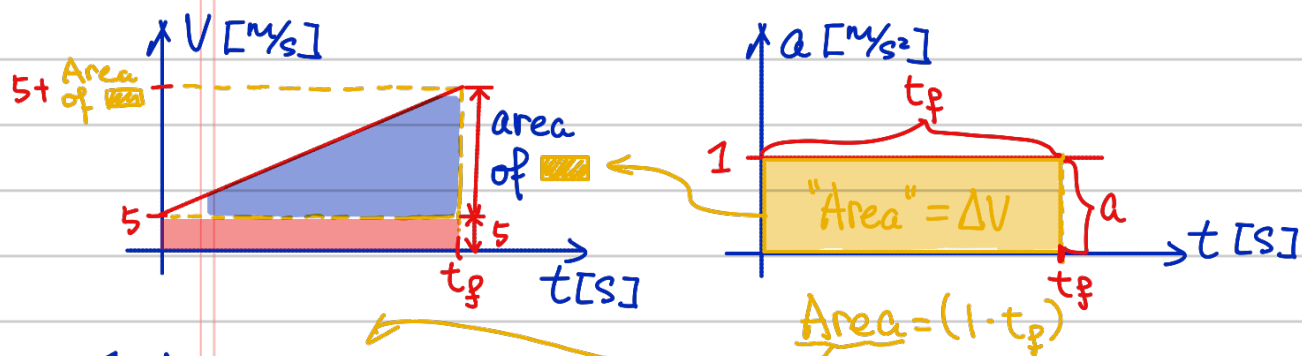
Step 6

For it to travel 200 m on the ramp. set $\Delta x'_{0 \rightarrow ?} = 200 \text{ m}$

$$\Delta x'_{0 \rightarrow ?} = \text{"Area" under } v \text{ from } 0 \text{ s to } ? \text{ s} = 200 \text{ m}$$

$$= \text{Area of } \triangle + \text{Area of } \square$$

(see pictures on next page)



Area = $(1 \cdot t_f) \cdot t_f / 2$

Area = $5 \cdot t_f$

$\Delta x'_{0 \rightarrow t_f} = \text{"Area" under } V \text{ from } 0 \text{ s to } t_f \text{ s} = 200 \text{ m}$
 $= \text{Area of } \triangle + \text{Area of } \square$

$200 = (1 \cdot t_f) \cdot t_f / 2 + 5 \cdot t_f$

quadratic equation for t_f

Solve for t_f using quadratic formula

$t_f = \frac{5^2 \pm \sqrt{5^2 - 4(\frac{1}{2})(-200)}}{2 \cdot (\frac{1}{2})} = 15.6 \text{ s}$

keep only positive t

time it takes for it to go up 100 m vertically.

Step 7

$200 = (1 \cdot t) \cdot t / 2 + 5 \cdot t \leftarrow \text{eqn we get in step 6}$
 $x - x_0 = a t^2 / 2 + v_0 t \leftarrow \text{formula in step 7}$

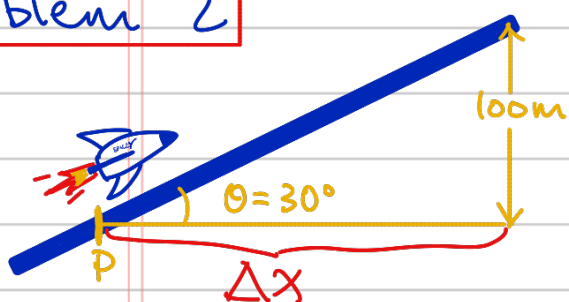
We have just solved for this quadratic eqn. $t = 15.6 \text{ s}$.

* This is where the formula comes from.

* In the future, if you see similar problem again with constant acceleration, apply the formula directly, no need to calculate the "areas" again.

Problem 2

trigonometry !!



$\tan(30^\circ) = \frac{100 \text{ m}}{\Delta x}$

$\Rightarrow \Delta x = \frac{100 \text{ m}}{\tan(30^\circ)} = 100\sqrt{3} \text{ m}$